

Online Appendix

비윤리적 친조직행동은 단일한가: 조직동일시의 선택적 관계와 윤리적 리더십의 조직 유형별 차이

Is Unethical Pro-Organizational Behavior Unitary? How its Associations with Organizational Identification and Ethical Leadership Vary by Behavior and Sector

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This document contains Tables A-F and Fig. A referred to in the article.

Appendix

Table A. Item-Level Multigroup Results

Predictor	Dependent variable	Public		Private		Sector difference	
		b (SE)	β	b (SE)	β	W	p
Organizational identification	1. Distorting facts	0.288*** (0.060)	0.201	0.216** (0.069)	0.142	0.63	0.427
	2. Exaggerating performance	0.307*** (0.062)	0.209	0.221** (0.072)	0.138	0.82	0.365
	3. Withholding negative information	0.286*** (0.063)	0.198	0.335*** (0.069)	0.223	0.28	0.597
	4. Recommending a low performer to leave	0.111 (0.065)	0.076	0.083 (0.067)	0.056	0.09	0.763
	5. Withholding service-error information from users	0.078 (0.063)	0.055	0.065 (0.067)	0.041	0.02	0.892
Ethical leadership	1. Distorting facts	0.015 (0.048)	0.012	0.225*** (0.058)	0.178	7.75	0.005**
	2. Exaggerating performance	-0.022 (0.050)	-0.018	0.314*** (0.060)	0.237	18.81	<0.001***
	3. Withholding negative information	-0.002 (0.050)	-0.001	0.209*** (0.058)	0.168	7.58	0.006**
	4. Recommending a low performer to leave	0.003 (0.053)	0.003	0.226*** (0.056)	0.184	8.36	0.004**
	5. Withholding service-error information from users	0.041 (0.049)	0.033	0.304*** (0.056)	0.233	12.58	<0.001***

Notes: Separate multigroup models with each item as the dependent variable. Age, gender, education, rank, and tenure are controlled. Equality tests use $\Delta df = 1$

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B. Correspondence between the Ethical Leadership Items and the Original Scale

No.	Brown, Treviño and Harrison (2005) item	Code	Item in this study
1	Listens to what employees have to say	—	Not included
2	Disciplines employees who violate ethical standards	—	Not included
3	Conducts his her personal life in an ethical manner	—	Not included
4	Has the best interests of employees in mind	—	Not included
5	Makes fair and balanced decisions	11-1	My supervisor makes fair and balanced decisions in handling work
6	Can be trusted	11-2	My supervisor can be trusted and followed
7	Discusses business ethics or values with employees	—	Not included
8	Sets an example of how to do things the right way in terms of ethics	11-3	My supervisor sets an example of handling work properly from an ethical standpoint
9	Defines success not just by results but also the way that they are obtained	11-4	My supervisor values the process as well as the results when discussing performance
10	When making decisions, asks “what is the right thing to do?”	11-5	My supervisor makes decisions by considering what is the right thing to do

Notes: Ethical leadership is measured with the five items of the ten-item scale of Brown, Treviño and Harrison (2005) that were included in the survey (items 5, 6, 8, 9, and 10). The included items concern fair decision making, trust, ethical role modeling, emphasis on process, and consideration of what is right, so they mainly capture the supervisor defining what is right and setting an example. Items on enforcing ethical standards, such as disciplining violators (2) and communicating ethical values (7), items on the relational side toward employees, such as listening (1) and considering employees' interests (4), and the item on the supervisor's personal life (3) were not included. Generalizing the results to all dimensions of the original scale therefore requires caution. Survey items are the author's English translation of the Korean questionnaire

Table C. Specifications of the Robustness Checks

Item	Specification	Remarks
Sampling weights	Post-stratification weights applied	Checks the influence of differences in sample composition
Response style	Acquiescence proxy added as a control	Person mean of the 75 items other than the UPB, OI, and EL items. Few reverse-worded items, so it is not fully separable from actual evaluations of the organization

Item	Specification	Remarks
Scale composition	3 items: 20-1, 20-2, 20-3	Performance-information distortion items only
	4 items: 20-1, 20-2, 20-3, 20-5	Excludes the low-performer item; matches prior studies using the same data
	5 items: 20-1 to 20-5	All items
Control variables	Six specifications	No controls → demographic variables → organizational perception variables added sequentially
Rank	Normalized rank index excluded sector-specific rank dummies	The baseline model uses the normalized rank index
Clustered standard errors	Organization level (public 373, private 956 organizations)	Mean respondents per organization 2.71 (max 40) in public and 1.05 (max 6) in private organizations. Single-respondent organizations number 243 and 926, so the cluster structure is limited
Estimator	WLSMV re-estimation	Indicators treated as ordinal; thresholds and loadings constrained to be equal across sectors
Leadership measures	Charismatic leadership entered jointly	Latent correlation with ethical leadership is 0.863 overall, 0.838 in public, and 0.903 in private organizations; it exceeds 0.85 in the full sample and in private organizations. The average variance extracted is below the squared correlation, so the Fornell-Larcker criterion is not met. Excluded from the baseline model because the unique effect is not stably identified

Note: Analyses were conducted in R 4.4.0 with lavaan 0.6.21. The analysis variables have no missing values (2,020 complete responses), so no missing-data procedure was applied

Table D. Measurement Items and Factor Loadings

Construct	Item	Content	Loading
Unethical pro-organizational behavior (UPB) $\alpha = 0.827$ Umphress, Bingham and Mitchell (2010)	20-1	If it helps my organization, I would distort facts somewhat to make my organization look good	0.839
	20-2	If it helps my organization, I would exaggerate my organization's performance somewhat	0.864
	20-3	If it helps my organization, I would withhold negative information about performance	0.703
	20-4	If my organization needs it, I would recommend that a low-performing employee move to another organization	0.442
	20-5	If my organization needs it, I would not inform users about a service that was provided incorrectly	0.631

Construct	Item	Content	Loading
Organizational identification (OI) $\alpha = 0.828$ Mael and Ashforth (1992)	1-1	When someone criticizes my organization, it feels like a personal insult	0.659
	1-2	I am very interested in what others think about my organization	0.632
	1-3	When I talk about my organization, I usually say “we” rather than “the company” or “the office”	0.614
	1-4	My organization's successes are my successes	0.746
	1-5	When someone praises my organization, it feels like a personal compliment	0.843
	1-6	If a news story criticized my organization, I would feel embarrassed	0.500
Ethical leadership (EL) $\alpha = 0.928$ Brown, Treviño and Harrison (2005)	11-1	My supervisor makes fair and balanced decisions in handling work	0.821
	11-2	My supervisor can be trusted and followed	0.873
	11-3	My supervisor sets an example of handling work properly from an ethical standpoint	0.871
	11-4	My supervisor values the process as well as the results when discussing performance	0.820
	11-5	My supervisor makes decisions by considering what is the right thing to do	0.858

Note: Loadings are standardized coefficients, all significant at $p < 0.001$. Fit of the measurement model: $\chi^2 = 485.8$ (df=101), CFI=0.972, TLI=0.967, RMSEA=0.047, SRMR=0.034. Item wording is the author's English translation of the Korean questionnaire; item codes follow the questionnaire

Source: Hur et al. (2024), Korea Public-Private Organization Members Perception Survey, Korea Institute of Public Administration

Table E. Detailed Coefficients of the Robustness Checks

Specification	Organizational identification		Ethical leadership	
	Public	Private	Public	Private
Baseline model	0.235***	0.174***	-0.001	0.263***
Sampling weights applied	0.218***	0.185***	0.013	0.249***
Acquiescence controlled	0.100*	0.120**	-0.095*	0.162***
UPB 3 items (performance-information distortion)	0.214***	0.196***	0.019	0.224***
UPB 4 items (excluding low-performer item)	0.185***	0.174***	0.028	0.241***
UPB 5 items (all)	0.177***	0.167***	0.027	0.249***
Organization-clustered standard	0.235***	0.174***	-0.001	0.263***

Specification	Organizational identification		Ethical leadership	
	Public	Private	Public	Private
errors				
WLSMV re-estimation (ordinal indicators)	0.220***	0.192***	-0.005	0.217***
Control: Rank excluded	0.176***	0.170***	0.026	0.249***
Control: Rank dummies	0.178***	0.173***	0.029	0.249***
Controls: none	0.135***	0.159***	0.026	0.255***
Controls: age				
gender	0.175***	0.166***	0.025	0.251***
Controls: + education, rank, tenure	0.177***	0.167***	0.027	0.249***
Controls: + subjective status, red tape	0.158***	0.147***	0.024	0.252***
Controls: + PSM				
job satisfaction	0.138***	0.135***	-0.020	0.233***
Controls: + autonomy				
goal clarity	0.115**	0.122**	-0.049	0.224***
fairness				
Charismatic leadership entered jointly	0.214***	0.179***	-0.172*	0.067

Notes: Cells are standardized coefficients. The baseline model controls for age, gender, education, rank, and tenure. Clustering adjusts standard errors at the organization level (1329 clusters)

Acquiescence is controlled by adding the person mean of the 75 items other than the key-variable items. The WLSMV and charismatic-leadership rows report the performance-information distortion model

The latent correlation between ethical and charismatic leadership is 0.863 overall, 0.838 in public, and 0.903 in private organizations; it exceeds 0.85 in the full sample and in private organizations, and the average variance extracted is below the squared correlation (Fornell and Larcker, 1981), so coefficients from this specification should be interpreted with caution

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table F. Direct Contrasts and 95% Confidence Intervals by Specification

Contrast	Specification	Difference (SE)	95% CI	Wald	p
OI, behavior difference (public)	Baseline MLR	0.199*** (0.053)	[0.096, 0.301]	14.30	<0.001
	Clustered SE	0.199*** (0.053)	[0.094, 0.303]	13.95	<0.001
	Ordinal WLSMV	0.261*** (0.043)	[0.177, 0.345]	37.44	<0.001
	Charismatic entered jointly	0.194*** (0.052)	[0.091, 0.296]	13.65	<0.001
OI, behavior difference (private)	Baseline MLR	0.182*** (0.051)	[0.083, 0.281]	12.98	<0.001
	Clustered SE	0.182*** (0.050)	[0.084, 0.281]	13.21	<0.001
	Ordinal WLSMV	0.230*** (0.042)	[0.148, 0.311]	30.49	<0.001

Contrast	Specification	Difference (SE)	95% CI	Wald	p
EL, sector difference (performance–information distortion)	Charismatic entered jointly	0.177*** (0.051)	[0.076, 0.278]	11.81	<0.001
	Baseline MLR	0.252*** (0.067)	[0.120, 0.385]	14.01	<0.001
	Clustered SE	0.252*** (0.072)	[0.111, 0.394]	12.28	<0.001
	Ordinal WLSMV	0.241*** (0.051)	[0.141, 0.341]	22.33	<0.001
	Charismatic entered jointly	0.264 (0.144)	[−0.019, 0.547]	3.34	0.068
EL, sector difference (third-party-directed conduct)	Baseline MLR	0.243*** (0.061)	[0.123, 0.362]	15.80	<0.001
	Clustered SE	0.243*** (0.060)	[0.124, 0.361]	16.16	<0.001
	Ordinal WLSMV	0.185*** (0.037)	[0.112, 0.258]	24.66	<0.001
	Charismatic entered jointly	0.269* (0.137)	[0.002, 0.537]	3.89	0.049

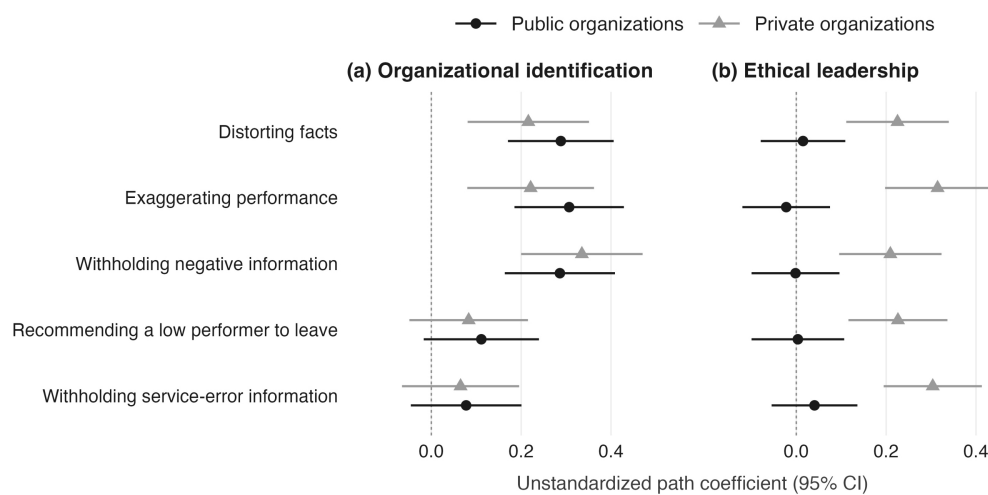
Notes: Differences are estimated directly as defined parameters in lavaan. The OI behavior difference is the coefficient for acceptance of performance–information distortion minus the coefficient for acceptance of third-party-directed conduct

The EL sector difference is the private–sector coefficient minus the public–sector coefficient

The baseline MLR, clustered, and charismatic specifications use observed means as dependent variables, whereas the ordinal WLSMV specification uses latent variables with the UPB items as ordinal indicators. Because the scales differ, the magnitudes of the contrasts are not compared across specifications

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Fig. A. Item–Level Path Coefficients



Note: Points are unstandardized path coefficients and lines are 95% confidence intervals